Executive Analytics Capstone

Customer Churn Analysis - Decision Brief

Business question: Which customer groups should a fictional subscription team contact first to reduce avoidable churn?

Scope: 15 supplied customer records. Findings are descriptive and should not be interpreted as causal or as a forecast for the full customer population.

1. Executive Snapshot

Metric	Observed value
Customers analyzed	15
Customers marked as churned	7
Observed churn rate	46.7%
Missing cells	0
Duplicate rows	0

2. Five Decision-Relevant Insights

1. Overall churn is 46.7% in the sample. 7 of 15 customers are marked as churned. This is the baseline for the supplied sample only.

2. Month-to-Month customers show 100.0% observed churn. All 7 month-to-month customers are marked Yes for churn, while the 4 one-year and 4 two-year customers are each at 0.0%. Because the groups are small, this is a directional signal rather than a population estimate.

3. Basic customers show the highest subscription-level churn in this sample. Basic: 5 of 7 churned (71.4%); Pro: 2 of 4 (50.0%); Enterprise: 0 of 4 (0.0%).

4. Higher support-ticket volume is strongly associated with churn in this sample. Customers with 4+ support tickets: 5 of 5 churned (100.0%); 2-3 tickets: 2 of 4 (50.0%); 0-1 tickets: 0 of 6 (0.0%). This does not show that support tickets cause churn.

5. Short-tenure customers have the highest observed churn. 0-6 months: 4 of 4 churned (100.0%); 7-12 months: 2 of 2 (100.0%); 13-24 months: 1 of 5 (20.0%); 25+ months: 0 of 4 (0.0%).

Contract Type Evidence

Group	Customers	Churned	Churn Rate
Month-to-Month	7	7	100.0%
One Year	4	0	0.0%
Two Year	4	0	0.0%

Subscription Type Evidence

Group	Customers	Churned	Churn Rate
Basic	7	5	71.4%
Enterprise	4	0	0.0%
Pro	4	2	50.0%

Tenure Group Evidence

Group	Customers	Churned	Churn Rate

0-6 months	4	4	100.0%
7-12 months	2	2	100.0%
13-24 months	5	1	20.0%
25+ months	4	0	0.0%

Support Ticket Evidence

Group	Customers	Churned	Churn Rate
0-1 tickets	6	0	0.0%
2-3 tickets	4	2	50.0%
4+ tickets	5	5	100.0%

3. Recommended Action

Audience: the fictional subscription retention team.
Decision: use short-tenure, month-to-month, Basic, and high-support-ticket customers as the starting groups for human retention review.
Supporting metrics: observed churn rates and subgroup counts above.
Next measurement: run the same segmentation on a larger customer population and compare churn before and after a defined retention pilot.

The recommendation is for review and testing, not automatic customer-level decisions. The overlapping group signals also mean the same customer may appear in several review groups.

4. Illustrative Impact Range (Scenario, Not Forecast)

If a future retention pilot prevented **1 to 2** of the 7 observed churns in this 15-customer sample, the observed churn rate would move from 46.7% to 40.0% or 33.3%. This is only an arithmetic scenario to show how impact should be quantified; it is not a prediction of real-world savings.

Assumptions for a real business impact estimate: define the eligible customer population, baseline churn rate, expected change from the intervention, and measurement period. Use historical data or a controlled pilot to test the assumptions.

5. Limitations and Data Quality

- The supplied sample contains only 15 customers.

- Small subgroup counts make percentages unstable; one customer can materially change a rate.

- The analysis is observational and descriptive. Correlation does not prove causation.

- The file has 0 missing cells and 0 duplicate rows, but the supplied sample may still differ from the wider customer population.

- Only the supplied fields were analyzed; other factors may affect churn.

Data source: EdVyro customer_churn_sample.csv supplied for the Executive Analytics Capstone.